

Ushtrime ND2ë

1.1.28 Guri është hedhur horizontalisht nga lartësia $h=2m$ mbi Tokë dhe ka rënë në distancën $s=7m$ prej vendit të rënies (duke llogaritur horizontalisht). Gjeni shpejtësinë fillestare v_0 dhe shpejtësinë e fundit v .

Zgjidhje:

Të dhënat:

$$h=2m$$

$$s=7m$$

$$v_0=?$$

$$v=?$$

$$\left\{ \begin{array}{l} v = v_0 + at \\ s = v_0 t + \frac{1}{2} at^2 \end{array} \right\} \begin{array}{l} s=h \\ a=g \end{array}$$

$$v = v_0 + gt$$

$$h = v_0 t + \frac{1}{2} gt^2$$

$$\left. \begin{array}{l} h=h' \\ v_0=0 \end{array} \right\} \begin{array}{l} h' = \cancel{v_0 t} + \frac{1}{2} gt^2 \\ h' = \frac{1}{2} gt^2 \cdot \frac{2}{g} \rightarrow \frac{2h'}{g} = t^2 / \sqrt{} \rightarrow t = \sqrt{\frac{2h'}{g}} \end{array}$$

$$v_0 = \frac{s}{t} = \frac{s}{\sqrt{\frac{2h'}{g}}} = s \cdot \sqrt{\frac{g}{2h'}} = 7m \sqrt{\frac{9,81 \frac{m}{s^2}}{2 \cdot 2 \frac{m}{1}}} = 7m \sqrt{2,4525 \frac{1}{s^2}}$$

$$v_0 = 7 \frac{m}{s} \cdot 1,566 = 10,96 \frac{m}{s} \approx 11 \frac{m}{s}$$

$$v = v_0 + gt \Rightarrow t = \frac{v - v_0}{g}$$

$$h = v_0 t + \frac{1}{2} gt^2$$

$$h = v_0 \left(\frac{v - v_0}{g} \right) + \frac{1}{2} g \left(\frac{v - v_0}{g} \right)^2$$

$$h = \frac{v_0 v - v_0^2}{g} + \frac{\cancel{g}(v^2 - 2vv_0 + v_0^2)}{2g\cancel{g}}$$

$$h = \frac{\cancel{2v_0 v} - \cancel{v_0^2} + v^2 - \cancel{v_0 v} + v_0^2}{2g}$$

$$h = \frac{v^2 - v_0^2}{2g}$$

$$2gh = v^2 - v_0^2 \rightarrow v^2 = v_0^2 + 2gh / \sqrt{}$$

$$v = \sqrt{v_0^2 + 2gh}$$

$$v = \sqrt{\left(11 \frac{m}{s}\right)^2 + 2 \cdot 9,81 \frac{m}{s^2} \cdot 2m}$$

$$v = \sqrt{121 \frac{m^2}{s^2} + 39,24 \frac{m^2}{s^2}} = \sqrt{160,24 \frac{m^2}{s^2}} = 12,65 \frac{m}{s}$$

$$v = 12,65 \frac{m}{s}$$

